

4. Activity Lifecycle Demonstration App

Topics covered: Activity Lifecycle, Log, Toast, state changes

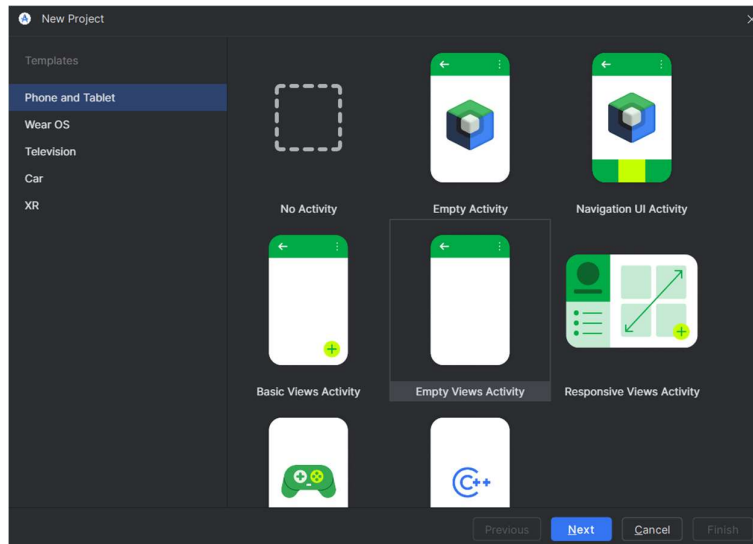
Task:

- Create an Activity
 - Override onCreate, onStart, onResume, onPause, onStop, onDestroy
 - Show Log messages for each method
 - Rotate device to observe lifecycle calls
-

Step 1:

Go to

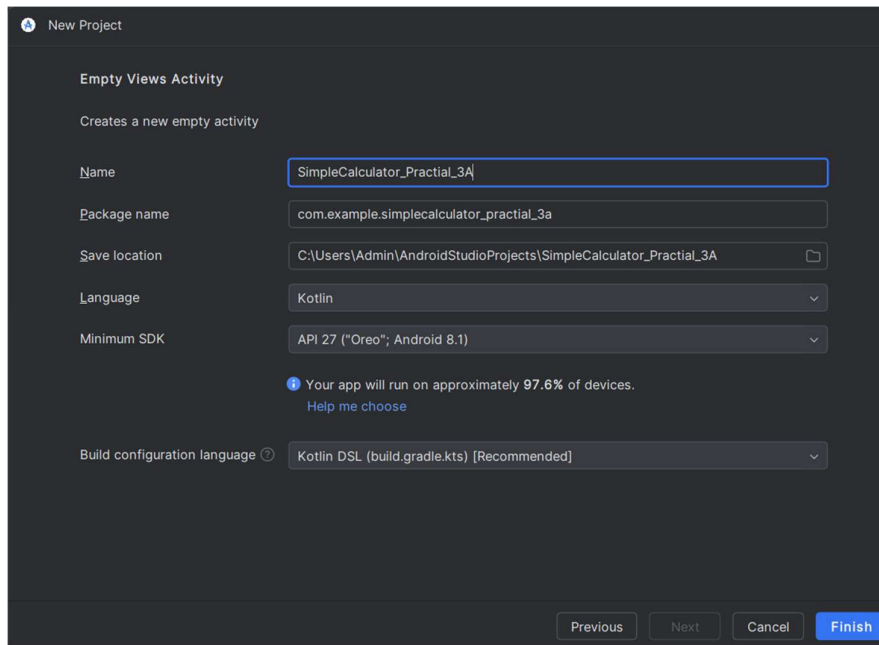
File → New → New Project → Empty Views Activity → Next



Step 2:

Click on **Project name** as

“LifecycleApp_Practical_4” → Finish



Step 3: Open the activity_main.xml

1. Go to
app → src → main → res → layout → activity_main.xml
2. Click it.
3. You will see **Design / Code / Split** tabs at the top.
4. Click **Split** (to see both code + preview)
or press **Alt + Shift + Left**
5. You will see XML code.

Replace it with the following code:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Activity Lifecycle Demo"
        android:textSize="20sp"
        android:layout_centerInParent="true"/>
</RelativeLayout>
```

Step 4: Open MainActivity.kt

1. Go to
app → src → main → java → com.example.lifecycleapp_practical_4 → MainActivity.kt
2. Open it.
3. Add these imports at the top:
4. Add the import at first:

```
import android.util.Log
import android.widget.Toast
```

Step 5: Add Lifecycle Code

After the existing `onCreate()` code (after `setContentView()`), **add and override the following methods:**

Add this code:

```
class MainActivity : AppCompatActivity() {

    private val TAG = "LifecycleDemo"

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        Log.d(TAG, "onCreate called")
        Toast.makeText(this, "onCreate()", Toast.LENGTH_SHORT).show()
    }

    override fun onStart() {
        super.onStart()
        Log.d(TAG, "onStart called")
        Toast.makeText(this, "onStart()", Toast.LENGTH_SHORT).show()
    }

    override fun onResume() {
        super.onResume()
        Log.d(TAG, "onResume called")
        Toast.makeText(this, "onResume()", Toast.LENGTH_SHORT).show()
    }

    override fun onPause() {
        super.onPause()
        Log.d(TAG, "onPause called")
        Toast.makeText(this, "onPause()", Toast.LENGTH_SHORT).show()
    }

    override fun onStop() {
        super.onStop()
        Log.d(TAG, "onStop called")
        Toast.makeText(this, "onStop()", Toast.LENGTH_SHORT).show()
    }
}
```

```
override fun onDestroy() {  
    super.onDestroy()  
    Log.d(TAG, "onDestroy called")  
    Toast.makeText(this, "onDestroy()", Toast.LENGTH_SHORT).show()  
}
```

Output (Virtual Phone):

Step 6: Run the App on Virtual Phone (Emulator)

1. Click the green RUN  button.
2. App will launch on emulator.
3. Toast messages will appear for:
 - onCreate
 - onStart
 - onResume

Step 7: Rotate the Device (Important Step)

1. Rotate emulator:
 - Portrait → Landscape
OR
 - Landscape → Portrait
2. Observe the lifecycle calls in this order:

onPause()

onStop()
onDestroy()
onCreate()
onStart()
onResume()

Step 8: Observe Logcat

- 1. Open Logcat**
- 2. Filter by tag:**

LifecycleDemo

- 3. You will see log messages for all lifecycle methods.**
-

Step 9: Connect App to Actual Phone (Same as Practical 2)

- 1. Open your phone → Settings**
- 2. Go to About device**
- 3. Tap Build Number 7 times**
→ “You are now a developer!”
- 4. Go back → Additional Settings**
- 5. Open Developer Options**
- 6. Enable USB Debugging**
- 7. Connect phone to laptop using USB cable**
- 8. Allow USB debugging permission**
- 9. Select your phone from device dropdown**
- 10. Click RUN **

Output (Real phone):

